

Generation Interconnection System Impact Study Report

For Queue Project Q20-1567

119Th Street 100 kV

March 04, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1567. The Interconnection Customer proposes a 181.8 MW battery energy storage facility interconnecting at the 119Th Street 100 kV substation in the Ohio area. The request was received on September 09, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 502 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 34 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$71,071,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fort Garland / Briarcliff Substation 100kV	Imperial / Eufaula 500kV	266.2	138	47.7
119Th St / Briarcliff 100kV	Eufaula / Imperial 500kV	255.1	132	47.7
Ft Garland / Beach Grove Substation 100kV L	Imperial / Eufaula 500kV	184.0	146	17.2
Marcos / Cleveland Thermal 100kV	Scranton / Marcos 100kV	171.3	130	8.0
Fowles / Wellington 1 100kV L	Eufaula / Imperial 500kV	167.0	134	5.9
Fashion / Getty 100kV	Getty / Fashion 100kV Line 1	163.8	157	11.4
Fashion / Abbe 100kV Line 2	Fashion / Abbe Substation 100kV Line 1	161.2	179	12.5
W Union / Wellington 1 100kV	Eufaula Substation / Imperial Substation 500kV	155.6	120	5.9
Getty / Fashion Substation 100kV line	Getty / Fashion 100kV #2	155.1	165	11.4
Fahion / Abbe 100kV	Fashion / Abbe 100kV	152.1	190	12.5
Scranton / Marcos 100kV L	Marcos / Cleveland Therml 100kV	148.9	130	6.7
Venture / Sierraville 100kV L	Sierraville / Venture 100kV	130.8	147	5.1
Sierraville / Venture 100kV Line 2	Venture Substation / Sierraville 100kV #1	130.8	147	5.1
Nrdonia / Hills 100kV	Chamberlin / Imperial 230kV	124.8	157	5.2
Getty / Ix Ctr 100kV	Ft Garland / Briarcliff 100kV line	124.8	168	7.5
Abbe / Gallatin Tn 100kV Circuit 1	Gallatin Tn / Abbe 100kV Cir. 2	124.5	162	5.1
Gallatin Tn / Abbe 100kV	Abbe / Gallatin Tn 100kV Circuit 1	124.5	162	5.1
Gallatin Tn / Abbe 100kV	Abbe / Gallatin Tn 100kV Cir. 1	124.5	162	5.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Poust / Fox 100kV L	Eufaula / Imperial 500kV	117.6	132	15.6
Joso / Pipeline 100kV Cir. 2	Joso / Pipeline 100kV Circuit 1	116.3	159	8.6
Stockton Hydro / Niland 100kV Line	Galeville / Marcos 100kV Line	115.7	133	18.1
Joso / Pipeline 100kV Circuit 1	Joso / Pipeline 100kV #2	114.7	161	8.6
Velasco / Ix Ctr 100kV	Ft Garland / Briarcliff 100kV	113.3	129	10.5
Hilltop Tap / Beach Grove 100kV	Bech Grove / Hilltop Tap 100kV Ckt 2	111.9	158	11.5
Sno-King / Pipeline 100kV Cir. 2	Sno-King / Pipeline 100kV Cir. 1	111.7	150	8.8
Hilltop Tap / Joso 100kV Ckt 1	Hilltop Tp / Joso 100kV Ckt 2	109.1	175	11.5
Hilltop Tap / Joso 100kV Circuit 2	Hilltop Tap / Joso 100kV	107.8	177	11.5
Velasco / Cedar Hill 100kV	Briarcliff / Ft Garland 100kV	107.0	124	10.5
Unionville / Beaver 230kV	Beaver / Eufaua 500kV	106.6	593	5.7
Sno-King / Pipeline 100kV	Sno-King / Pipeline 100kV Circuit 2	105.6	159	8.8
John P Madgett / Dollarhide 100kV line	Eufaula / Beaver 500kV	105.5	247	5.8
Marcos / Gleville 100kV line	Imperial / Eufaula 500kV line	103.9	297	12.0
Sno-King / Greene County Station 100kV Ckt 1	Greene County Sta / Sno-King 100kV Circuit 2	102.9	146	9.0
Fox / Niland 100kV	Galeville Substation / Marcos 100kV line	102.1	123	18.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ft Garland / Briarcliff 100kV L	Rebuild line with higher-capacity conductor	\$2,878,000	100% (\$2,878,000)
119Th St / Briarcliff 100kV	Rebuild line with higher-capacity conductor	\$1,217,000	100% (\$1,217,000)
Beach Grove / Ft Garland 100kV	Rebuild line with higher-capacity conductor	\$3,894,000	100% (\$3,894,000)
Cleveland Thermal / Marcos 100kV Line	Rebuild line with higher-capacity conductor	\$1,443,000	52% (\$757,000)
Fowles / Wellington 1 100kV L	Rebuild line with higher-capacity conductor	\$2,841,000	100% (\$2,841,000)
Fashion / Getty 100kV	Rebuild line with higher-capacity conductor	\$4,809,000	100% (\$4,809,000)
Abbe / Fashion Substation 100kV #2	Rebuild line with higher-capacity conductor	\$2,211,000	100% (\$2,211,000)
W Union / Wellington 1 100kV	Rebuild line with higher-capacity conductor	\$2,396,000	100% (\$2,396,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Getty / Fashion Substation 100kV line	Rebuild line with higher-capacity conductor	\$4,168,000	100% (\$4,168,000)
Abbe / Fashion 100kV Line 1	Rebuild line with higher-capacity conductor	\$3,523,000	100% (\$3,523,000)
Scranton / Marcos 100kV L	Rebuild line with higher-capacity conductor	\$2,806,000	80% (\$2,257,000)
Sierraville / Venture 100kV	Rebuild line with higher-capacity conductor	\$1,756,000	100% (\$1,756,000)
Sierraville / Venture 100kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,364,000	100% (\$1,364,000)
Nordonia / Hills 100kV	Rebuild line with higher-capacity conductor	\$3,828,000	11% (\$427,000)
Getty / Ix Center 100kV line	Rebuild line with higher-capacity conductor	\$2,550,000	100% (\$2,550,000)
Abbe / Gallatin Tn 100kV Circuit 1	Rebuild line with higher-capacity conductor	\$3,206,000	100% (\$3,206,000)
Gallatin Tn / Abbe 100kV Ckt 2	Rebuild line with higher-capacity conductor	\$2,763,000	100% (\$2,763,000)
Gallatin Tn / Abbe 100kV	Rebuild line with higher-capacity conductor	\$2,204,000	100% (\$2,204,000)
Poust / Fox 100kV L	Rebuild line with higher-capacity conductor	\$3,831,000	100% (\$3,831,000)
Joso / Pipeline 100kV Cir. 2	Rebuild line with higher-capacity conductor	\$3,169,000	100% (\$3,169,000)
Stockton Hydro / Niland 100kV Line	Rebuild line with higher-capacity conductor	\$1,167,000	100% (\$1,167,000)
Joso / Pipeline Substation 100kV L	Rebuild line with higher-capacity conductor	\$2,798,000	100% (\$2,798,000)
Ix Ctr Substation / Velasco 100kV	Rebuild line with higher-capacity conductor	\$2,068,000	100% (\$2,068,000)
Hilltop Tap / Beach Grove 100kV	Rebuild line with higher-capacity conductor	\$1,440,000	100% (\$1,440,000)
Sno-King / Pipeline 100kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,976,000	100% (\$2,976,000)
Hilltop Tap / Joso 100kV Ckt 1	Reconductor line	\$826,000	100% (\$826,000)
Joso / Hilltop Tap 100kV Line 2	Reconductor line	\$606,000	100% (\$606,000)
Velasco / Cedar Hill 100kV	Reconductor line	\$1,070,000	100% (\$1,070,000)
Unionville / Beaver 230kV	Reconductor line	\$4,450,000	40% (\$1,783,000)
Sno-King / Pipeline 100kV Cir. 1	Reconductor line	\$875,000	100% (\$875,000)
John P Madgett / Dolarhide 100kV	Reconductor line	\$575,000	53% (\$302,000)
Galeville / Marcos 100kV	Reconductor line	\$2,341,000	88% (\$2,071,000)
Sno-King / Greene County Station 100kV Ckt 1	Reconductor line	\$380,000	100% (\$380,000)
Fox / Niland 100kV L	Reconductor line	\$488,000	100% (\$488,000)
	Total allocated to project		\$71,071,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.